

Squiggle: Multimodal Lasso Selection in the Real World

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Figure 1: Left: An illustration of a user performing a lasso selection with Squiggle. The green and orange markings are to illustrate hand and gaze behavior, but would not be visible to the user. Right: Three examples of Squiggle lassos drawn by users in our data collection, placed over the target objects based on the users' gaze locations while drawing (Sec. 5.0.1).

Abstract

Smart glasses devices are emerging with egocentric cameras and gaze tracking, raising the possibility of new interaction techniques that enable users to reference real-world objects they wish to digitally interact with. However, many of these devices lack a display, making precise object referencing difficult due to the lack of continuous visual feedback. We introduce Squiggle, an interaction technique that enables users to reference real-world objects without continuous feedback by drawing an invisible loop or “lasso” with an

imagined ray-cast pointer. Through a virtual reality data collection, we observed that this gesture can elicit useful gaze behavior in addition to providing drawing input itself. Based on these results we implemented and evaluated a real-world prototype of Squiggle, demonstrating that it can improve accuracy of object referencing over Gaze + Pinch alone, particularly for selecting compound objects and groups.

CCS Concepts

• **Human-centered computing** → **Gestural input**; **Mobile devices**; **User studies**.

Keywords

Smart Glasses, Eye Tracking, Segment Anything, Lasso Selection



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1 Introduction

A new class of smart glasses are emerging with egocentric cameras and gaze tracking capabilities, but without visual displays [25, 27]. These systems provide a compelling platform for digital interactions with the physical world because they can map the user's gaze to objects in their environment. For example, in a grocery store a user may look at a product and request information on it, or at home a user sorting books may place their gaze on one book in particular and add it to a "donate" list. However, simply mapping a single gaze point to the real world may be insufficient for referring to multiple objects (e.g., a stack of books), or to a specific part of a complex scene (e.g., a brand logo on a package). Many screen-based applications offer lasso selection (i.e., drawing a loop enclosing the items of interest) for these types of use cases, but this technique relies on continuous visual feedback which is not possible on display-free devices.

To enable more precise referencing of real-world objects without visual feedback, we introduce Squiggle, a multi-modal object referencing technique for smart glasses in concert with a wrist- or hand-worn device capable of sensing hand angles and pinch gestures. Squiggle uses the familiar metaphor of lasso selection with a laser pointer, but without visual feedback. The user performs a rough lasso-drawing input to indicate what they are referencing, and this input is synthesized with the natural gaze behavior they exhibit while drawing the lasso. Through developing and evaluating Squiggle, this work explores the usefulness of these input signals with the research question "Can invisible lasso-drawing enable more accurate referencing of real-world objects than gaze pointing alone, either through the lasso-drawing input itself or the natural gaze behavior elicited by the act of drawing the lasso?"

To answer this, we conducted a data collection (n=12) in which participants were asked to select items in several virtual scenes, covering a range of visual sizes, distances, and object hierarchy positions (e.g., part of something larger, compound object, group of objects). They performed these selections using our midair lasso technique at two levels of detail ("Draw a Loop" and "Draw an Outline") as well as with a baseline Gaze + Pinch technique involving only a single point of gaze input.

Our analysis of this data found that Squiggle's lasso gesture was useful in two ways. First, it provided a drawing shape whose dimensions could be used to predict the rough size of the user's intended target. Second, the gesture often caused users to perform a series of gaze fixation points following the contour of their target.

We compared five candidate methods for converting Squiggle's gaze and drawing input into queries for an image segmentation model. Based on the results and the data we had collected, we produced two fine-tuned versions of EfficientSAM [61] (a lightweight segmentation model): one for our gaze-only baseline and one for our lasso-drawing technique. We incorporated these models into

a prototype system for an image masking task using display-free smart glasses [9] and a sensor-enabled ring [20].

We evaluated this prototype with a user study (n=12) in which participants were asked to select a variety of targets with both the baseline and the lasso technique, and finally to select targets using a "hybrid" technique in which they could freely choose between the two forms of input for each selection. Our results found that Squiggle's lasso input enabled users to reference groups of objects and compound objects with greater accuracy than the Gaze + Pinch technique, and it was chosen more often by participants for these kinds of targets. Participants appreciated the hybrid technique's flexibility to support both expressive lasso selections and the lower-friction Gaze + Pinch selections.

This paper contributes:

- A novel design and implementation for expressive in-world object referencing with no display, inspired by screen-based lasso selection techniques.
- Findings on user gaze and gesture behavior during display-free lasso selection, from a virtual reality data collection.
- Evaluation results demonstrating improved accuracy for selections of compound objects and groups of objects with our lasso prototype compared to a Gaze + Pinch technique.

2 Related Work

This work proposes a solution to the in-world referencing problem, i.e. communicating an object or objects of interest in one's physical environment to a digital system. It builds on previous explorations of multiple-object selection techniques, as well as research into natural gaze behavior as a tool for input.

2.1 In-World Referencing

There are many reasons one may want to refer a digital system to something in their environment, such as to annotate it, save it for later, or inquire about it. There is a large body of work exploring ways to perform such references naturally and with convenient sensors. Some of these involve pointing at a target with one's hand, as if to refer another person to it. [21, 45, 66]. Nimble [45] predicted the user's object of focus using the position of their index fingertip in a camera image. LookHere [66] predicted saliency in an image based on deictic gestures such as pointing or holding up something. WorldPoint [21] used cameras and LiDAR on a mobile phone to predict the target of a user's pointing gesture, and achieved a pointing error rate of 15.1 cm across ranges of 0.8 to 2.4 m. Squiggle also involves a type of midair pointing gesture (i.e., with an imaginary laser pointer), though we are not interested in the accuracy of this pointer's direction; instead we use gaze tracking to predict the target's location.

In augmented reality (AR) and smart glasses systems, gaze data is often combined with an egocentric camera image to determine an object of interest [1, 25, 27, 37, 54]. Museum Guide 2.0 [54] tracked users' gaze as they explored a mock museum, and provided additional information about "exhibits" when the user's gaze dwelled over them. Hales et al. [15] used hand poses to send commands to smart objects when the user's gaze was over an object's marking tag. ubiGaze [1] detected when participants' gaze fixated on a target object, and enabled them to leave a virtual message attached to the

object for a companion to receive later. These works demonstrate compelling use cases for in-world referencing, but they could be improved with the addition of more expressive input. For example, someone in a museum could specifically request information about an object in a painting, rather than the painting itself. Or someone shopping could annotate a whole set of dishware, rather than a specific plate.

More recently, several works have explored combining gaze and camera data with a voice query to a virtual assistant [25, 27, 37]. For example, GazePointAR [27] replaced pronouns like “that” in a user’s query with the name of the recognized object nearest to their gaze location. GazeGPT [25] sent an image of the user’s view along with their query to the AI agent. Noting that the user’s intended precision could not necessarily be gathered from their gaze input, GazeGPT made and sent three images with every query, cropped to different potential scales around the user’s gaze point. Combined with the context of the user’s voice query, GPT-4V was often able to correctly deduce the intended target. However, voice input is not always desirable, and may still be ambiguous in some cases, so there remains a need for expressive nonverbal referencing.

Shi et al. [50] investigated region selection in augmented reality through specifying two opposite corner points for a 2D bounding box. While this kind of input could be used in similar scenarios to those Squiggle is designed for, it was not evaluated for in-world referencing tasks; participants only copied rectangles given by the researchers. We also prefer lassos to bounding box input for in-world referencing because lassos support a wider range of shapes, and they may be easier for users to imagine when there is no display to indicate a grid alignment for boxes.

2.2 Multiple-Object Selection

Selection of multiple elements is a common task across many devices and applications. In screen-based interfaces, lasso selection (i.e. drawing a loop enclosing the elements one wishes to select) has been a staple for decades, and has been modified or adapted in several ways. For example, ICE-Lasso [7] predicted the cluster of elements that a user was trying to select as they drew, allowing them to stop early if it was correct. Harpoon Selection [28] and Lazy Selection [62] reduced the amount of drawing users would need to perform by having them draw directly through their targets rather than around them. The present work leverages users’ familiarity with lasso selection to enable referencing of physical objects in the user’s environment, and differs from past implementations of the technique in that the user’s drawing is not visible to them.

Some works have explored multiple-object selection in 3D scenes [4, 14, 17, 53, 60, 65, 67]. Lucas [17] evaluated techniques for multiple-object selection in VR, including a lasso selection (implemented through a 2D floating tablet view of the scene that users could draw on) that participants preferred to placing a 3D bounding cube. Stenholt et al. [53] compared several methods of selecting large numbers of small objects in virtual reality, and found that a “magic wand” which selects automatically from a single point and a spherical brush tool both worked well under different circumstances, compared to a 3D box, which users didn’t prefer. Tangible Brush [4] explored volumetric selection by drawing a lasso on a tablet and then extruding it by moving the tablet through physical space, and

Gosset et al. [14] later implemented this in VR. LassoNet [67] introduced a deep neural net for selecting relevant subsets of 3D point clouds based on a 2D lasso drawing. Though Squiggle was designed for use in real-world (i.e., 3D) environments, it differs from most VR techniques in that its input is not a three-dimensional volume. Like LassoNet, it uses a two-dimensional drawing projected into the scene. It is also distinct in that it is for smart glasses that cannot provide visual feedback to aid with selection.

PinchCatcher [22] recently explored holding a partial pinch gesture to enable a mode during which the user could gaze at and subselect multiple objects to be included in their selection. This interaction involves many similar input modalities to Squiggle (i.e., gaze tracking, pinch gestures, and arm movements), but it focuses on serial sub-selection while Squiggle targets simultaneous grouping. PinchCatcher’s approach is well-suited for precise selection of discrete virtual objects, such as an arbitrary subset of files in a folder. However, it may be less desirable for our smart glasses use case, as it could be difficult to keep track of many subselections without feedback, and the boundaries between “objects” in the real world aren’t always clear [63].

2.2.1 Image Segmentation. When interacting with images of real-world scenes, the set of select-able elements is less well-defined than in many digital interfaces. Potential targets may completely include each other (e.g. a logo on a shirt, the whole shirt, the person in the shirt), so determining from a user’s input exactly what portion of the scene they intend to reference is a difficult problem. There has been a great deal of research on the task of image segmentation, i.e. intelligently “cutting out” a mask of an object or objects from an image based on user queries. Though this was once done with algorithms based on features like colors and edges [31, 46], now machine learning models like the Segment Anything Model (SAM) [23] can be trained to produce masks of objects in images based on simple user prompts, such as a few points or a bounding box. SEEM [68] introduced a novel decoding mechanism for unifying a wide range of prompts, like other images or scribbles on an image. Semantic-SAM [29] generated a set of masks at several potential granularity levels for any given prompt. SAM meets Gaze [2] and GazeSAM [57] explored screen-based image segmentation tasks using gaze tracking to produce prompts. Squiggle’s prototype uses EfficientSAM [61], a lightweight and easily-trainable version of SAM, to segment images of a user’s surroundings with a prompt derived from gaze and gesture data from drawing a midair lasso.

2.3 Eye Gaze as Input

There is a large body of HCI work related to using one’s gaze direction like a cursor to point at UI elements [18, 22, 26, 40, 41, 43, 48, 56, 58]. A popular implementation of this involves using a pinch gesture to trigger selection of the object under the user’s gaze. Velloso et al. [56] found that this approach was faster than hands alone for selecting and dragging a virtual object on a computer screen. Pfeuffer et al. [41] explored several applications of this selection technique in virtual reality and coined the term “Gaze + Pinch,” which we use throughout this paper. Mutasim et al. [40] compared pinch as a selection trigger against gaze dwell and clicking a button, and validated that pinching was a reasonable alternative to

button clicks. Gaze + Pinch is now used in commercial mixed reality devices such as the Apple Vision Pro¹, and we use a similar interaction in the present work as our state-of-the-art baseline.

Some of these techniques also incorporate a pinch-and-hold or pinch-and-drag gesture for the purpose of manipulating an object or cursor after selection [41, 56]. Squiggle’s drawing gesture is distinct from these in that it is entirely part of the initial selection input. We do not place and move a cursor in the image plane; instead we use ray-drawing (with the hand) to elicit natural gaze behavior, which is then combined with the drawing shape to provide a richer referencing signal.

Previous research has investigated ways to harness gaze tracking as input by utilizing its natural movement patterns. VRPursuits [19], Outline Pursuits [52], and Vergence Matching [51] all explored different ways to determine a user’s intended target by smoothly moving key virtual items and identifying which one’s trajectory the user’s gaze was following. Other approaches rely on gaze behavior that users may not be aware of at all, i.e. that people naturally look ahead to pointing targets [5, 35, 49, 64]. MAGIC pointing [64] intelligently warped a user’s cursor to near the location of their gaze when it moved a long distance. GazeRayCursor [5] enhanced the accuracy of VR raycasting by estimating the target’s depth from the user’s gaze direction. EyePointing [49] and Gaze-Hand Alignment [35] both triggered a selection when the user’s arm-pointing ray [49], fingertip [35], or cursor [35] crossed the item under their gaze. Squiggle applies a similar idea by treating the user’s natural gaze behavior as a primary input while they are performing an arm gesture, though in this case that gesture is drawing a lasso rather than pointing.

3 Technique Design

Squiggle was designed for smart glasses capable of tracking gaze direction and capturing egocentric images, but with no visual display [9, 25]. Such systems are becoming increasingly popular in the form of smart glasses with AI capabilities², and we expect they will remain a viable alternative to head-mounted AR displays due to their comparatively low weight and cost, and support for unobtrusive interactions. For gesture input, we assume a hand-worn device capable of tracking hand orientation and thumb-to-forefinger pinches. In this work we used a smart ring [20] but these sensing capabilities are available on commercial smart watches³ and wristband research systems [8, 13]. We decided not to pursue head-worn hand tracking due to low reliability of tracking and pinch detection, which has been shown to cause frustration for users [40, 41].

Given this technology, and the task of referring to an object in the physical world, we consider the Gaze + Pinch technique [41] to be a state-of-the-art baseline. Though it’s normally used for virtual reality selections, we adapted it for in-world referencing in the following way: users invoke a selection by tapping their thumb to their index finger, at which point their current gaze direction and egocentric view are captured and used to determine what object(s) they are referring to. This is a similar flow to the world-referencing input used by GazeGPT [25], though without an accompanying

voice query. A single gaze point on the world is relatively quick and easy for users to produce, but we expected that one point would be insufficient to cover the full range of complex and sometimes multiple-object referencing needs that may arise. Our goal was to create an expressive referencing technique that could exist alongside Gaze + Pinch as an optional mode for scenarios that would benefit from richer input.

We adapted a familiar and expressive multiple-object selection technique (i.e. lasso selection) for in-world referencing, with the following interaction design: at a high level, a Squiggle user selects a target by sweeping an imaginary laser pointer in a lasso around it (Fig. 1). We chose a hand posture where the user’s thumb is up and their palm faces inward, similar to how one might hold a remote control or laser pointer. In the interest of comfort and avoiding occlusion, users do not need to lift their arm into their field of view, similar to the posture used for Gaze&Hand [35]. The user touches their thumb to the side of their index finger to begin “drawing” (though their input is invisible to them), moves their hand to draw the appropriate lasso around their target, and finally lifts their thumb from their finger to stop drawing and submit the input for processing. As they draw, their gaze direction is also being recorded as part of the input. The user may provide as much or as little detail as they wish, from a loop to a more complex outline. For simple selection targets they may still invoke the classic Gaze + Pinch by tapping their thumb to their index finger and immediately releasing it.

4 Data Collection

Before implementing a prototype of Squiggle, we wanted to better understand how users would perform this kind of referencing input. We conducted a user study with the following goals: a) Observe and analyze patterns in users’ behavior while making Squiggle-style selections. b) Collect data we could use to develop a model for predicting target objects from Squiggle input. Though our ultimate goal was to design an interaction technique for referencing objects in the real world, we conducted this data collection in virtual reality so we could incorporate a wider range of environments and targets than we could keep controlled in the real world (e.g. a city street corner). In VR we were also able to generate a perfect “ground truth” image mask for every sample, which was important to our analysis and modeling.

4.1 Apparatus

We conducted the study using a Meta Quest Pro headset⁴ as the display and for vision-based gaze tracking. Participants also wore a ring-based input device similar to Electroring [20] on their dominant index finger for rapid and precise thumb-to-finger touch detection (using active electrical bioimpedance sensing) and orientation tracking (using an inertial measurement unit, or IMU).

4.2 Participants

Twelve participants were recruited to participate in the study (3 male, 8 female, 1 non-binary, ages 23 - 59, all right-handed). They reported a mean familiarity with AR and VR of 2.4 out of 5.

¹<https://www.apple.com/apple-vision-pro/>

²<https://www.meta.com/ai-glasses/>

³<https://www.apple.com/watch/>

⁴<https://www.meta.com/quest/quest-pro/>



Figure 2: (a) A user in our study environment, wearing the Meta Quest Pro and ring input device as they make a selection. (b) Four examples of scenes in our study from the user's perspective, accompanied by details about a few example selection targets found in that scene.

4.3 Task

Participants were asked to select a series of virtual target objects using one of three techniques. In increasing expressiveness, these techniques were:

- (1) *Gaze + Pinch.* Our baseline technique. The user looks at their target and pinches to trigger a selection.
- (2) *Draw Loop.* The user performs a quick lasso selection, pointing and sweeping an imaginary laser in a loop around their target.
- (3) *Draw Outline.* The user performs a more careful lasso selection, outlining the rough shape of their target with their imaginary laser.

They were not given any kind of feedback about their input during or after their selection, except the researcher's verbal guidance to select the next target.

4.3.1 Targets. All of our 65 selection targets came from one of the following Unity asset packs⁵⁶⁷ and were found in virtual scenes resembling a busy city street, a residential street, or an apartment. We varied the visual size of targets (ranging from 0.2 degrees squared to 1802.2 degrees squared from the user's perspective) and their distance from the user (ranging from 1.2 m to 73.2 m from the user's head).

To cover scenarios that we anticipated would benefit from lasso input, we included targets that were parts of larger objects, targets that were compound objects, and targets that were groups of objects, in addition to standard single-object targets. It can be difficult to define what counts as a single object, so we had three raters assign "hierarchy labels" to the targets. First, they could mark targets

as being a "part" of a larger object in the scene or not. For this label our raters had a Fleiss' Kappa [10] agreement score of 0.90. Independently from that label, targets could be marked as "groups" of multiple smaller objects, "compound objects" (i.e., having sub-parts that could stand alone as objects, while still being one object), or neither. For these labels our raters had a Fleiss' Kappa agreement score of 0.84. Taking the majority opinion in cases of disagreement, and using the "compound" or "group" label over "part" in cases where both were applied, 30 of our 65 targets were given the "part" label, 7 were given the "group" label, and 7 were given the "compound objects" label, leaving 21 targets for the label "simple object." Examples of these can be found in Figure 2b.

4.4 Procedure

The study consisted of three back-to-back sessions with brief (i.e., less than five minutes) breaks in between, lasting approximately two hours total. Each session was dedicated to a single technique from Gaze + Pinch, Draw Loop, and Draw Outline, whose order was counterbalanced across participants. At the start of a session, participants would be given instructions for the technique they were going to perform, then walked through the Quest Pro's built-in gaze tracking calibration and a ring calibration in which they lined up their hand in a thumb-on-top posture with a mid-air laser to demonstrate what angle felt like "pointing forward" to them. After this the ray was turned invisible for all trials, but the user's hand was still rendered in light gray in the virtual scene throughout the study.

Our selection targets were organized into "batches" of 6-10 that were all to be selected from the same virtual location. For each session the order of batches was randomized, and the order of targets within each batch was randomized. At the start of each batch, the participant would be placed in the appropriate virtual location

⁵<https://assetstore.unity.com/packages/3d/environments/urban/anime-tokyo-japanese-city-258316>

⁶<https://assetstore.unity.com/packages/3d/props/interior/smith-s-home-2-36495>

⁷<https://assetstore.unity.com/packages/3d/environments/modern-supermarket-186122>

and taught about the targets they would be selecting there. A researcher would highlight the targets in bright green one at a time while saying their name (e.g., “This is the banana.”), and confirm that the participant understood before moving on. After this teaching period, the researcher would walk the participant through the selection trials with verbal instructions only (e.g., “Please select the banana.”) and no visible changes in the virtual environment, to simulate a real world interaction.

4.5 Measures

During and around users’ selections, we recorded their hand’s pinch state and orientation at 60Hz. From the change in orientation after a pinch begins, we derived a 2D spherical drawing path in degrees. We also recorded the user’s view of the virtual scene at around 12 FPS, their gaze direction at 60Hz, and their head movement at 60Hz, which we used to project gaze data into camera images from other points in time. Notably, we did not use the participant’s 3D pointing ray, because this would be more difficult than orientation to track for our smart glasses use case. It also likely would not help to locate the target in space, as people’s ability to point accurately in 3D is worse than state of the art gaze tracking [6, 9, 11, 21, 32, 38, 39].

4.6 Hypotheses

We had the following hypotheses about how users would behave with Squiggle:

- (1) *Users’ drawing size would relate to their target’s size.* We expected to find a correlation between the angular height and width of the user’s target with that of the shape they drew when making a selection. Such a relationship could be useful to disambiguate how much a user intends to select.
- (2) *The act of drawing would evoke useful natural gaze data from users.* Users tend to look at their target while performing pointing interactions [30, 35, 58, 64], and when tracing a path with their finger they usually fixate their gaze at points of high curvature on the path [12]. We expected that a user’s gaze may follow the contour of their target while they try to draw around it, even if the drawing is imaginary.

4.7 Results

4.7.1 Post-processing. For our analyses we found it helpful to perform a few post-processing steps. We chose to work with only one camera frame per selection, the frame from nearest to the start of the user’s pinch gesture, in order to keep the measurements simple. Additional frames from the same selection would be mostly redundant, because users stood still during all selections. We identified gaze fixations using an I-DT algorithm [34], and used those instead of our raw 60 Hz gaze samples. For some Loop and Outline selections, we observed that the user’s gaze would drift away from the target as they were finishing their drawing, so we removed up to 15% of fixations from the end if they were moving away from the starting gaze point. An example of this post-processing can be seen in Fig. 3a. We also defined the concept of a “trimmed drawing,” where we identified the point of least distance between the first third and the last third of a user’s drawing stroke (often an intersection where the primary lasso was closed), and kept only the drawing that occurred between those points. We used trimmed

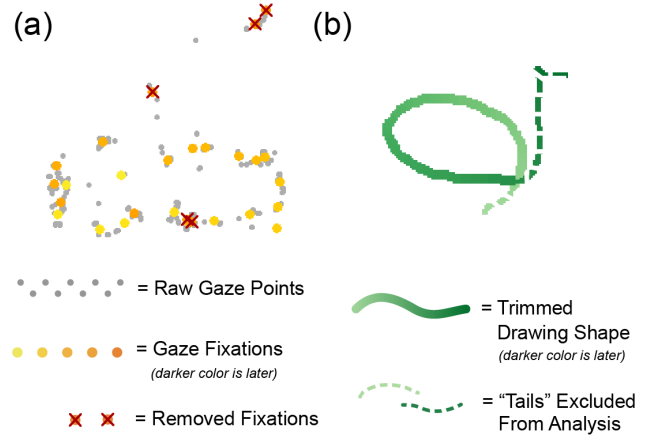


Figure 3: (a) Example of our post-processing on the gaze data from a selection. Fixation points are extracted from the raw gaze data, and some may be trimmed from the end if they drift away from the rest. (b) Example of our post-processing on the drawing data from a selection. “Tails” trailing away from the main loop are identified and excluded from the drawing’s bounding box.

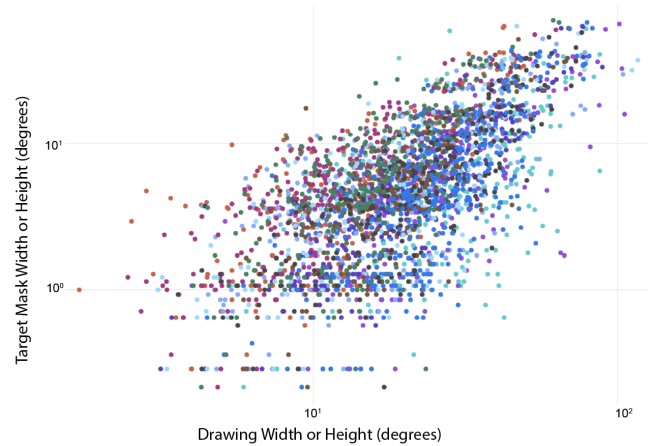


Figure 4: Scatter plot of users’ drawing height or width vs their target’s height or width (both in degrees), for both drawing techniques. There is a roughly linear relationship between them (slope = 0.46, $R^2 = 0.49$). Each participant’s samples are plotted in a different color. Both axes use logarithmic scale, as this is close to the distribution of target heights and widths in our study.

drawings when analyzing a drawing’s height, width, and midpoint, to reduce the impact of long “tails” at the start or end that are not part of the main shape. An example of this can be seen in Fig. 3b.

4.7.2 Drawing Dimensions. We compared the heights and widths (in degrees) of each target object with that of the trimmed drawing the user made to select it. We looked at height and width rather than area because the same area can be produced with many very

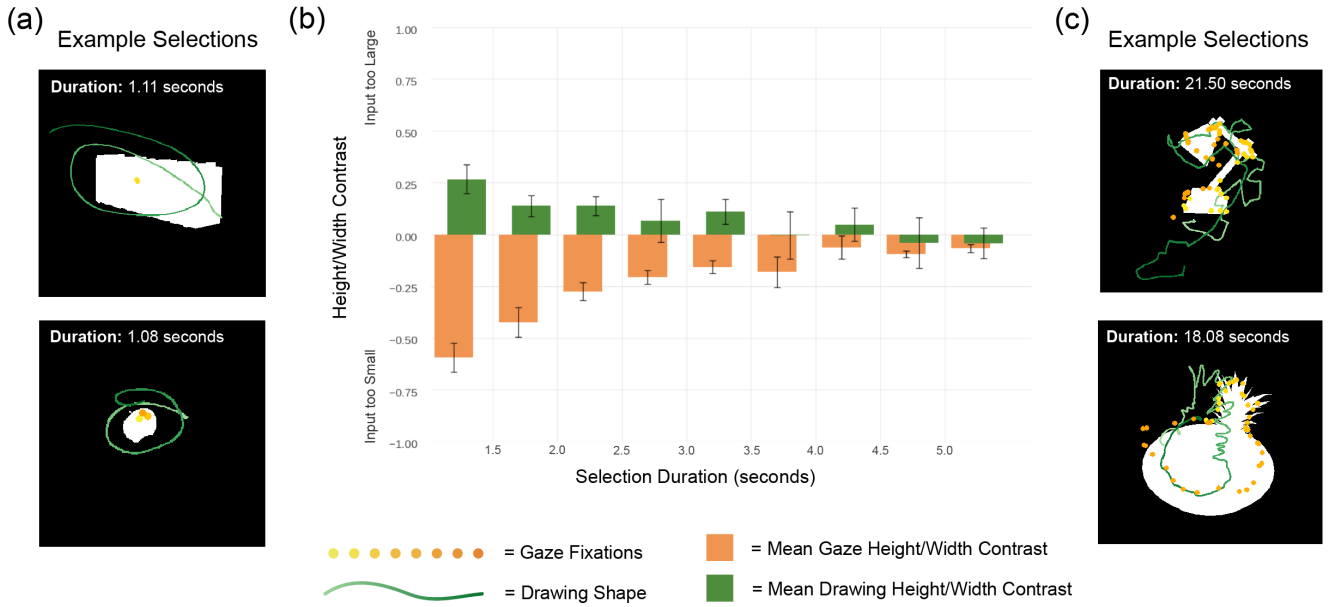


Figure 5: (a) Examples of quick, “cursory,” lasso selections where the user’s gaze is relatively stationary and their drawing conveys the target’s dimensions. The user’s drawing has been scaled and placed over their target mask (rendered in white) using the approach in Section 5.0.1. (b) The mean contrast (proportional difference) between the dimensions of the user’s fixation points and that of the target, as well as the contrast between the dimensions of the user’s post-processed drawing and that of the target. (c) Examples of slow, “careful” lasso selections, where the user’s gaze traces around their target as they draw.

different aspect ratios. Therefore, we judged height and width to be a better measure of a lasso’s size accuracy. These height and width values, including those from both the Draw Loop and Draw Outline techniques, are plotted in Figure 4. A linear regression of these values resulted in the equation $L_t = 0.460L_d - 2.308$, where L_t is the height or width of the target and L_d is the respective height or width of the user’s trimmed drawing, with $p < .001$ for both coefficients. The R^2 value for this overall relationship was 0.490, though many participants showed a stronger correlation within just their own samples, with a median per-participant R^2 value of 0.625. These values indicate that, while there is significant noise, a user’s drawing dimensions can help roughly predict their intended target’s dimensions. This supports our first hypothesis. For all future analyses, we used “rescaled” drawing data, with the drawing’s height and width scaled according to the above linear relationship.

4.7.3 Gaze Behavior. We observed two high-level types of selection behavior for the drawing techniques, which we’ll call “careful” and “cursory,” roughly associated with slower and faster selections respectively. These each derive a different value from the drawing input. With careful selection behavior, users’ gaze would loosely trace around the edge of their target as they created their drawing (Fig. 5c). In these cases, the gaze data seems to be totally sufficient to determine their target’s location, shape, and size. The drawing data itself isn’t necessary at all, though it’s the task of drawing that resulted in this useful gaze behavior. These observations support our second hypothesis. This kind of behavior was exhibited by all participants when performing Draw Outline selections, and half of

the participants consistently behaved this way with Draw Loop as well.

On the other hand, six participants exhibited cursory selection behavior with the Draw Loop technique, in which their gaze would not move much as they swept their invisible laser around their target (Fig. 5a). In these cases, the gaze data is useful to determine the target’s location, but the drawing provides important signal as to the target’s dimensions. There was a spectrum of behaviors in between the two extremes, which we observed to be related to the amount of time spent performing the selection gesture. To better understand this relationship, we plotted the mean value of the contrast ($c = (v_1 - v_2) / (v_1 + v_2)$) between the target’s height or width and that of the user’s gaze fixations (Fig. 5b). We did this for each half-second bucket of selection durations from 1.0-1.5 seconds to 5.0-5.5 seconds, for the six participants who had selections in all of these ranges (i.e. those who performed cursory selections). We also plotted the mean contrast between the target’s dimensions and those of the scaled and trimmed drawing data. As selection time increases, we see that the user’s gaze fixations become an increasingly good representation of the target’s size. However, with brief selections (i.e. under about two seconds), the gaze’s spread is much smaller and less likely to be a useful predictor of the target’s size. Meanwhile, the drawing contrast changes less as a function of time, so it could be a more useful predictor of the target’s size than gaze at brief durations. We wanted Squiggle to support quick and cursory lasso selections like this, so we made sure to handle this behavior when developing our prototype.

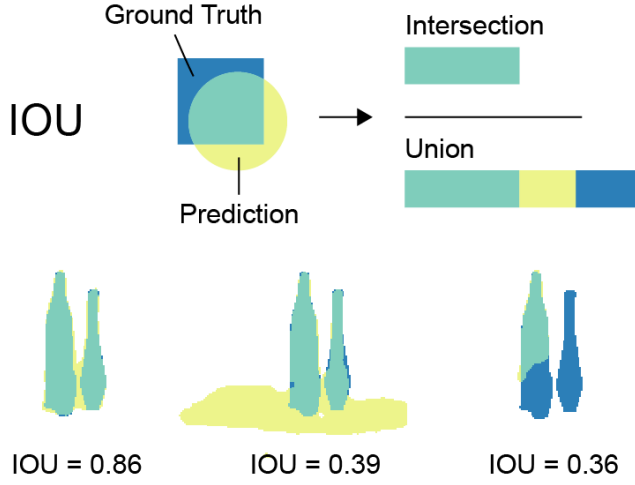


Figure 6: An illustration of the intersection over union (IOU) metric, with three examples from our SAM2 experiments.

5 Prototype Development

We used the results of our data analysis to inform development of an interaction technique for an in-world image segmentation task, i.e. a task where the user performs a selection gesture and the desired output is a 2D mask of their target in a camera image. We chose to use mask output because it preserves enough data to be useful for a wide range of referencing tasks. In practice, a system may often perform object identification or reverse image search on the user’s target, but we did not want the performance of these operations to influence our investigation. Another benefit of mask output is that it’s simple to objectively evaluate its accuracy: we measure intersection over union, or IOU, of the resulting mask with a ground truth mask of the target. An IOU of 1.0 would mean that the two masks are exactly identical, but we note that this is much higher precision than is needed for many tasks. Examples of IOUs for several masks are shown in Fig. 6.

5.0.1 Drawing Input. For our prototype’s input, we combined the Draw Loop and Draw Outline techniques from our data collection into a single lasso-drawing technique of variable detail. We decided to perform the same post-processing steps (i.e. fixation detection and drawing trimming) that we did in our data analysis (4.7.1), and apply our cumulative linear regression from Section 4.7.2 to scale all of the drawing data. To place each drawing on its respective camera image, we positioned it such that the center of the bounding box enclosing the post-processed drawing was aligned with the center of the bounding box enclosing the filtered gaze fixation points. This was based on the assumption that, when the user’s gaze spread was small, they would likely be gazing near the center of their target [3]. Note that for all visualizations of user drawings over their targets in this work (Fig. 1, Fig. 5), we scale and place the drawing using this method, and we show the full drawing including any “tails” that would be ignored for the purposes of determining scale and placement. After this pipeline was built, the next component we needed was a model that could produce an image mask from this lasso input.

5.1 Image Segmentation Models

We leveraged versions of the Segment Anything Model (SAM) [23] for image segmentation. SAM can efficiently produce a mask of an object given an image and a query, such as a set of points on the object or a box roughly enclosing the object. SAM has mostly been used in screen-based interfaces where the user can see the queries they are making, though recent research also showed potential in querying SAM with gaze data [2, 24, 57]. We experimented with different ways to query SAM2 (i.e., the latest and most state-of-the-art form of SAM) [44] using Squiggle data, and then fine-tuned two prompt encoders for EfficientSAM (i.e., the fastest and most easily-tuned version of SAM) [61] to be used in a real-time evaluation.

5.2 SAM2 Experiments

Before we would train our own prompt encoder for Squiggle input, we wanted to develop an intuition for the best ways to convert gaze and drawing input into queries for a Segment Anything Model. We compared the following five methods, illustrated in Figure 7:

- (1) *Mean Fixation, Gaze + Pinch Only*: For our baseline input, we query SAM2 with a single point, the mean gaze fixation point from the time when the user was pinching. Because the pinch is so brief for this technique, this is usually only one point. When there are no fixation points during the pinch, we use the most recent fixation before the pinch began.
- (2) *Gaze Bounding Box (GazeBB)*, Drawing Techniques Only: A bounding box that encloses all of the user’s gaze fixations during their selection. This method uses the gesture input only to mark the start and end times, and otherwise relies entirely on the user’s natural gaze behavior while drawing.
- (3) *Drawing Bounding Box (DrawBB)*, Drawing Techniques Only: A bounding box that is the height and width needed to enclose the user’s post-processed drawing, centered on the bounding box of their gaze fixations during the drawing.
- (4) *Max Bounding Box (MaxBB)*, Drawing Techniques Only: A bounding box centered on the user’s gaze bounding box, but with the height and width needed to enclose all gaze fixations and also the user’s post-processed drawing. In Fig. 5, the larger of the two input types often has the contrast value closer to zero, so we thought that taking the maximum of the two could improve overall accuracy.
- (5) *Interpolated Bounding Box (IntBB)*, Drawing Techniques Only: A bounding box centered on the user’s gaze bounding box, with a height (h) and width (w) determined by the duration (d) of their selection according to the following formula:

$$h_{IntBB} = \max\left(\frac{d}{4}, 1\right) h_{GazeBB} \cdot \left(1 - \max\left(\frac{d}{4}, 1\right)\right) h_{DrawBB}$$

$$w_{IntBB} = \max\left(\frac{d}{4}, 1\right) w_{GazeBB} \cdot \left(1 - \max\left(\frac{d}{4}, 1\right)\right) w_{DrawBB}$$

In other words, from zero to four seconds, we interpolate linearly between the dimensions of the drawing’s bounding box and those of the gaze fixations’ bounding box. Beyond four seconds, we use the gaze bounding box only. This approach was designed to weight gaze data more heavily for careful selections and drawing data more heavily for cursory selections (Section 4.7.3).

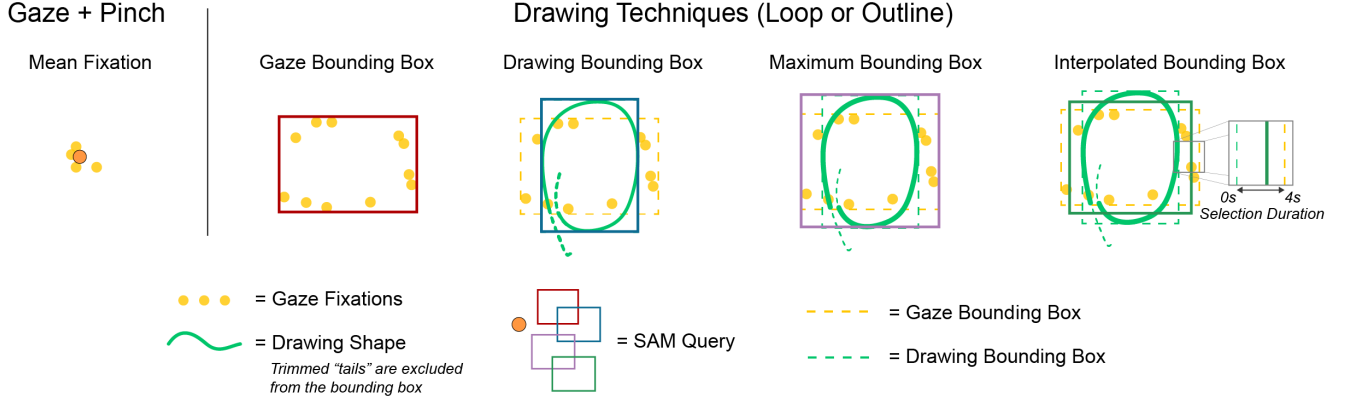


Figure 7: Illustrations of the five types of queries we tested with SAM 2: one single-point query for Gaze + Pinch and four bounding box queries for Drawing techniques.

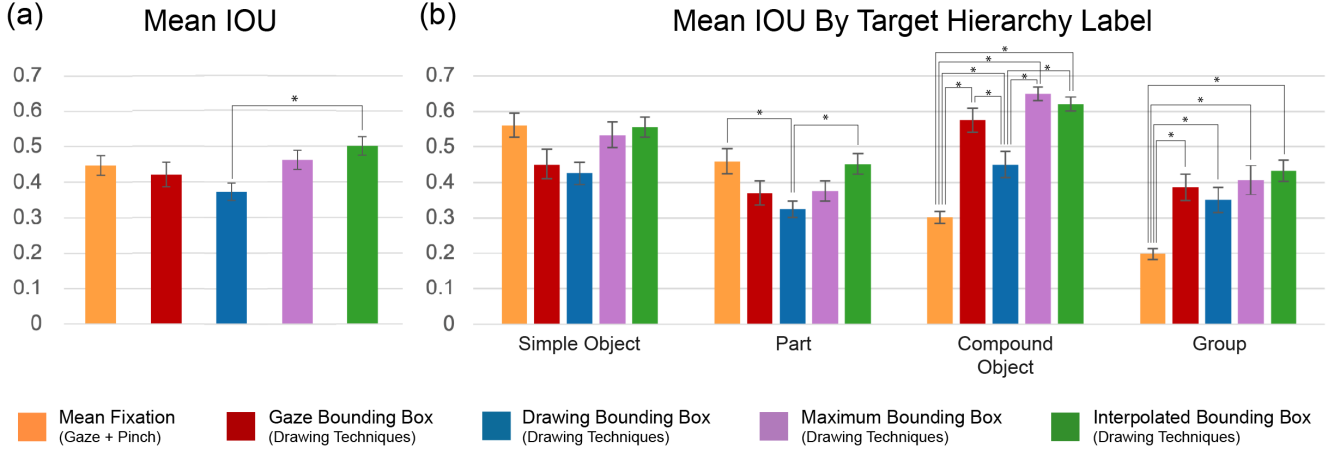


Figure 8: (a) Mean mask IOUs for each query method used in our exploration. Error bars represent standard error and * denotes a significant result ($p < 0.05$ after correction). (b) Mean mask IOUs for each query method used in our exploration, sorted by target hierarchy label. Error bars represent standard error and * denotes a significant result ($p < 0.05$ after correction).

We generated each of these queries for our recorded selections of the appropriate input type, prompted SAM2 with them, and evaluated the IOUs of the resulting masks and our ground truth.

5.2.1 Results. The results of this experiment are in Figure 8. An RM-ANOVA was conducted to analyze the mean IOU for each query method, overall as well as within each target hierarchy label. The results of these were:

- **All Targets:** $F(4, 44) = 4.91, p = .002$
- **Simple Objects:** $F(4, 44) = 5.93, p < .001$
- **Parts:** $F(4, 44) = 5.54, p = .001$
- **Compound Objects:** $F(4, 44) = 40.31, p < .001$
- **Groups:** $F(4, 44) = 17.73, p < .001$

We then conducted pairwise Tukey tests within each category. Our results found that the Interpolated Bounding Box (IntBB) queries had significantly higher mean IOUs than the Drawing Bounding Box (DrawBB) queries, both overall ($p = .024$) and for parts of objects ($p = .043$). Our baseline method, Mean Fixation, worked

relatively well in general, even outperforming DrawBB on parts of objects ($p = .0277$, post-hoc Tukey test), but it had significantly lower mean IOUs than every bounding box method for the label “compound object” ($p < .001$ for all four, post-hoc Tukey tests) and the label “group” ($p = .003$ for GazeBB, $p = .022$ for DrawBB, $p < .001$ for MaxBB, and $p < .001$ for IntBB, post-hoc Tukey tests). This is consistent with our initial assumptions about this baseline and our motivation for Squiggle. Though DrawBB often performed poorly itself, it’s worth noting that this drawing data was used to produce the bounding boxes for MaxBB and IntBB, which often performed the best.

Based on our experiments, we chose to use the interpolation approach from IntBB for lasso selections with our prototype. We then moved into the fine-tuning phase. We switched from using the state-of-the-art SAM2 to EfficientSAM [61] at this point, because EfficientSAM is a much lighter model tailored for on-device use cases like the smart glasses scenario for which we designed Squiggle.

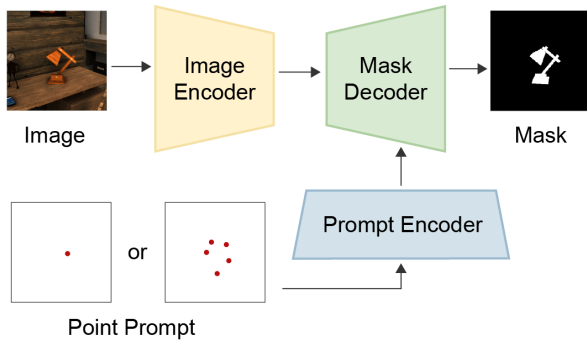


Figure 9: A diagram of the model architecture we used when fine-tuning EfficientSAM.

5.3 EfficientSAM Fine-Tuning

The EfficientSAM model consists of three primary components:

- The **Image Encoder** processes visual input.
- The **Prompt Encoder** encodes prompts or queries.
- The **Mask Decoder** generates object masks based on the encoded information.

To effectively integrate our query data into the EfficientSAM model, we fine-tuned the prompt encoder and mask decoder, while keeping the image encoder frozen. By doing so, we enabled the model to accommodate our distinct type of input while still leveraging the robust image understanding capabilities of the pre-trained image encoder. We fine-tuned two different prompt encoders, one for Gaze + Pinch and one for lasso selections. Each fine-tuning process used a training set of samples for the main model training, as well as disjoint validation and testing sets for selecting and evaluating the best models.

For the baseline Gaze + Pinch technique, we gathered four more participants' worth of data following the same procedure as our original data collection but for only Gaze + Pinch. This was done to bring our number of selections closer to that of our lasso-drawing techniques. Then, using all 16 total users' data (i.e., 1040 selections), we generated the Mean Fixation queries the same way we did for our SAM2 experiments and randomly selected 730 of these to be our training set, with the remaining 310 split between testing and validation sets (i.e., 155 each).

For our lasso technique, we used data from both the Draw Loop and Draw Outline sessions of our original twelve participants, a total of 1560 selections. For each selection we sampled 5 points over the length of the trimmed and scaled drawing path (centered on the gaze bounding box) and sampled 5 more points over the length of the path of the user's filtered gaze fixations. We interpolated between the pairs of points in corresponding positions using the same rule as for our Interpolated Bounding Box queries before: linear interpolation from the drawing point to the gaze point based on the selection duration, with selections longer than 4 seconds using only the gaze. This resulted in a set of 5 query points roughly encircling the target. We randomly selected 1092 of these to be our training set, with the remaining 468 split between testing and validation sets (i.e., 234 each).

Once fine-tuning was done, we added the two new models to our pipeline, connected our sensing devices, and implemented a hybrid mode that could flexibly apply either model. More information about these are in Sections 6.1 and 6.3. With this, our prototype was complete, and it was time to validate its real-world performance.

6 Evaluation Study

We conducted an evaluation study of our image segmentation prototype with the following goals: a) measure the accuracy of selections performed with the Lasso technique compared with Gaze + Pinch, b) evaluate the qualitative user experience of both of these techniques, c) assess the feasibility of a hybrid technique that allows users to choose between Gaze + Pinch and Lasso.

6.1 Apparatus

Participants wore Project Aria glasses [9] for egocentric image capture, gaze tracking, and the SLAM-based relative head tracking needed to project gaze data into earlier camera images. Participants also wore a ring-based input device similar to Electroring [20] on their dominant index finger for thumb-to-finger touch detection (using active electrical bioimpedance sensing) and orientation tracking (using an inertial measurement unit, or IMU).

6.2 Participants

Twelve new participants were recruited to participate in the study (4 male, 7 female, 1 non-binary, ages 24-47, all right-handed). They reported a mean familiarity with AR and VR of 3.3 out of 5.

6.3 Task

Participants were asked to perform selections of target objects in the room using one of three techniques. These techniques were:

- (1) *Gaze + Pinch*. Our baseline technique. The user looks at their target and pinches to trigger a selection.
- (2) *Lasso*. The user performs a lasso selection, pointing and sweeping an imaginary laser in a loop around their target.
- (3) *Hybrid*. The user performs a selection with their choice of either of the above techniques, and the system determines which one it was based on their thumb-to-finger touch duration. We used 0.5s as the cut-off duration beyond which the system would interpret the input as a Lasso, because none of our data collection drawing samples were shorter than that.

Participants were not given any kind of feedback during their selection. After the selection, the system would display on a computer screen the captured image with its top predicted target mask highlighted in yellow (Fig. 10b). When the technique was Lasso, or the Hybrid technique determined that the user performed a Lasso selection, it would also display the scaled drawing in green over the image. Participants would be asked to rate the accuracy of the mask before proceeding.

6.3.1 Targets. The study included 23 targets, all physical objects that we arranged in the study room. Their visual sizes ranged from 4.1 degrees squared to 877.4 degrees squared, and their distance from the user ranged from 1.0 m to 4.0 m.

Like our data collection, we included several targets that were parts of larger objects, compound objects, or groups of objects. To

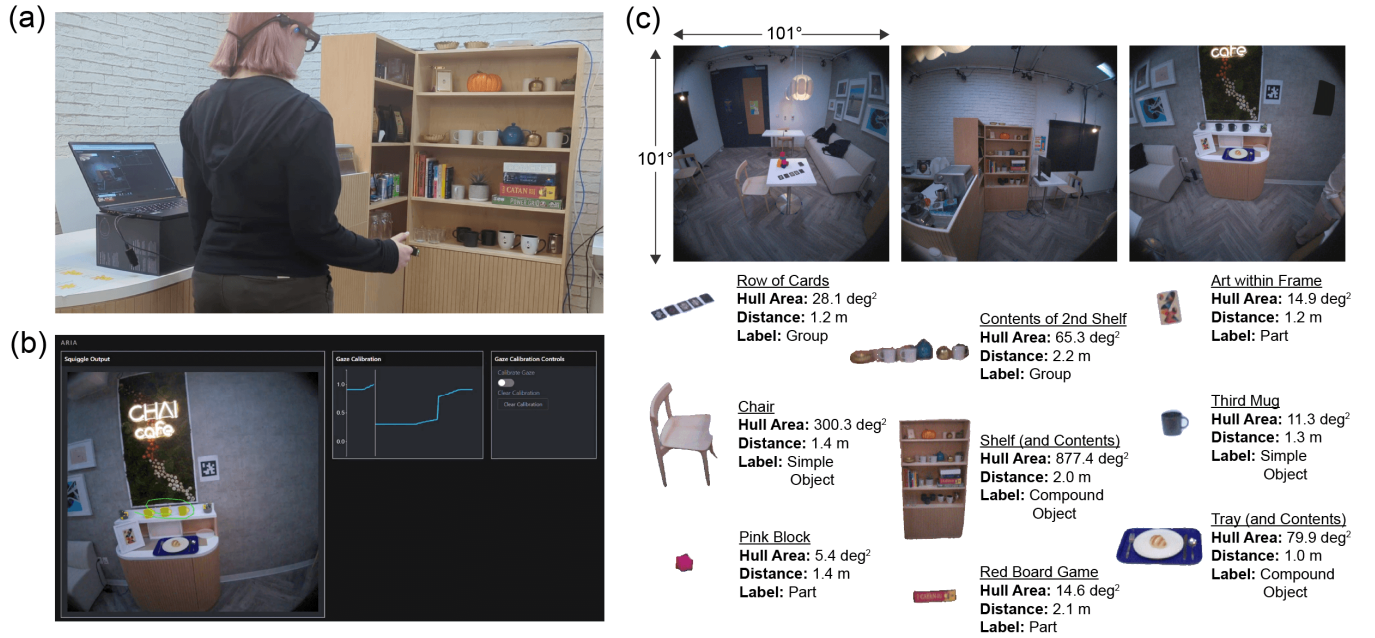


Figure 10: (a) A user in our study environment, wearing the Project Aria glasses and ring input device as they make a selection. (b) Screenshot of our study desktop interface, which displays the user’s adjusted drawing input in green and the system’s predicted mask in yellow over the camera image captured during the selection. (c) Several egocentric images captured during the study, demonstrating some of the targets used.

determine these categories we once again had three raters assign “hierarchy labels” to the targets. Our raters had a Fleiss’ Kappa [10] agreement score of 0.88 when choosing to mark targets as “parts” or not, and a score of 0.86 when choosing to mark targets as “compound objects,” “groups,” or neither. Ultimately 7 of our 23 targets were classified as parts, 6 were classified as compound objects, and 6 were classified as groups, leaving 4 simple objects. Examples of these can be found in Figure 10c.

6.4 Design and Procedure

A repeated-measures design with within-subjects factors was used. The independent variables were selection technique and target hierarchy label. The study consisted of three back-to-back sessions with brief (i.e., less than five minutes) breaks in between, lasting approximately ninety minutes total. Each session was dedicated to a single selection technique, where the first two sessions would be Gaze + Pinch and Lasso in a counterbalanced order, and the final session would be for the Hybrid technique, so that users could make decisions about which technique to use after having some experience with both.

At the start of each session, participants would be given instructions for the technique they were going to perform, then walked through calibration procedures for all devices. To calibrate “pointing forward” relative to the ring, participants pointed their index finger (with the ring on) at the opposite wall, imagined the resulting ray, and then adopted the thumb-on-top hand posture with their imaginary laser pointing along that ray.

After this they began selection trials. For each target, the researcher would name it, the participant would perform the selection, then the participant would look at the model’s output and provide a rating. The targets’ order was randomized for each session. At the end of each session, participants answered a portion of the NASA Task Load Index (TLX) Survey [16] about their experience using that technique. At the end of all three sessions, they filled out another qualitative survey comparing the different techniques.

6.5 Hypotheses

We had the following hypotheses:

- (1) *Lasso selection would result in higher average mask IOU than Gaze + Pinch for compound objects and groups.* This would align with what our SAM2 experiments suggested, and with the role of lasso selection traditionally.
- (2) *Gaze + Pinch would require lower physical effort than Lasso,* because the Gaze + Pinch gesture is briefer and users don’t need to lift their arm to perform it.
- (3) *In the Hybrid session, participants would choose to use Lasso for compound objects and groups, and choose Gaze + Pinch for other cases.* This would be consistent with them preferring to minimize physical effort, but also understanding that there is improved accuracy associated with Lasso for compound objects and groups.
- (4) *Participants would prefer the Hybrid technique,* because it enables them to balance the effort and accuracy trade-offs of the other two techniques.

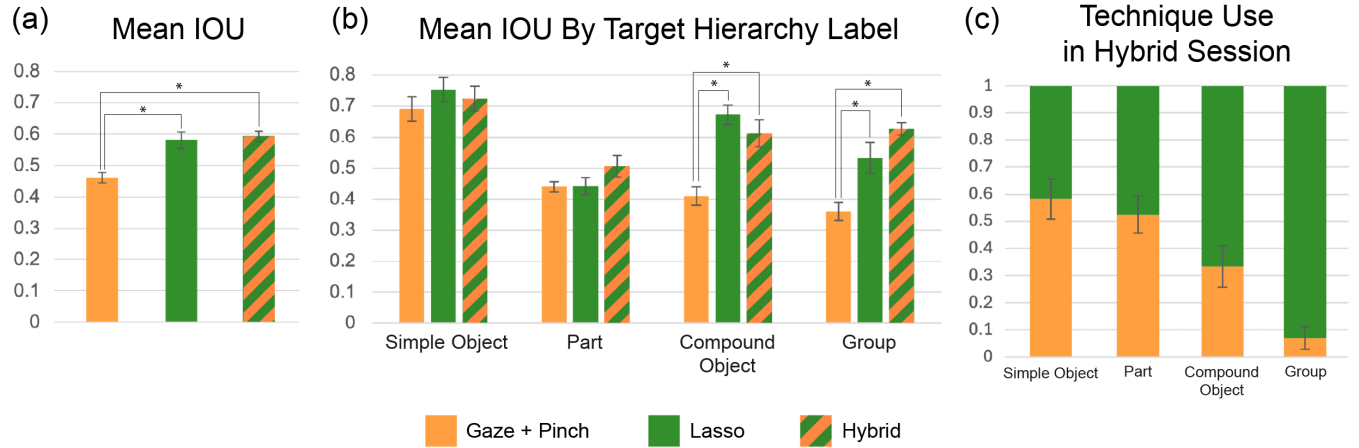


Figure 11: (a) Mean IOU (intersection over union) values for the Gaze + Pinch, Lasso, and Hybrid techniques. Error bars represent standard error and * indicates a significant ($p < 0.05$) result. (b) Mean IOU values for the selection techniques, sorted by target hierarchy label. Error bars represent standard error and * indicates a significant ($p < 0.05$) result. (c) Proportion of selections in the Hybrid session that participants chose to make with the Gaze + Pinch technique vs the Lasso technique, sorted by target hierarchy label. Error bars represent standard error.

6.6 Results

6.6.1 Mask Accuracy. An RM-ANOVA was conducted to analyze the mean IOU for each technique, overall as well as within each target hierarchy label. We then conducted pairwise Tukey tests within each category. The results of our RM-ANOVAs were:

- **All Targets:** $F(2, 22) = 14.98, p < .001$
- **Simple Objects:** $F(2, 22) = 1.02, p = .38$
- **Parts:** $F(2, 22) = 1.38, p = .27$
- **Compound Objects:** $F(2, 22) = 17.89, p < .001$
- **Groups:** $F(2, 22) = 22.764246, p < .001$

As shown in Figure 11a, the Lasso selection technique had an overall mean IOU of 0.58, a significant improvement over Gaze + Pinch’s mean IOU of 0.46 ($p < .001$, pairwise Tukey test). The Hybrid technique had a mean IOU of 0.59, also a significant improvement over Gaze + Pinch ($p < .001$, pairwise Tukey test). When broken down into categories based on object hierarchy label (Fig. 11b), no significant differences were found between the techniques for simple objects or parts, but Lasso and Hybrid both significantly outperformed Gaze + Pinch for referencing compound objects ($p < .001$ for Lasso, $p = .002$ for Hybrid, pairwise Tukey tests) and groups ($p = .006$ for Lasso, $p < .001$ for Hybrid, pairwise Tukey tests). Because participants always tried the Hybrid technique last, there could be learning effects in its performance. However, it only showed significant differences from Gaze + Pinch when the Lasso technique also did. We can draw our conclusions from only the comparison between Lasso and Gaze + Pinch. The results support our first hypothesis, and speak to the value of the Lasso technique, i.e. it provides increased accuracy for referencing some targets when compared with Gaze + Pinch.

6.6.2 Selection Effort. We were not able to verify our second hypothesis directly. The responses to the NASA TLX questions did

not differ significantly between conditions, except for the question “How successful were you at making selections with this technique?”, where Hybrid (mean 4.8 out of 7) outperformed Gaze + Pinch (mean 3.9 out of 7), with adjusted $p = .03$. Our full TLX results are included in supplementary materials.

We also analyzed the mean selection time for each technique, measured from thumb touch to thumb release (i.e., excluding potential planning or preparation time). On average, Gaze + Pinch selections took 0.34 seconds (SD 0.09s), Lasso selections took 2.27 seconds (SD 0.88s), and Hybrid selections took 1.42 seconds (SD 0.52). These results aren’t surprising given the techniques’ respective designs, but they highlight a benefit of Gaze + Pinch input, i.e. it’s very quick. This could be a reason for participants to choose Gaze + Pinch when referencing simple objects.

6.6.3 Hybrid Technique Behavior. Our simple 0.5s threshold for distinguishing Gaze + Pinch selections from Lasso selections worked in most cases, with 6 total errors across 276 Hybrid selections (2.17%). Five of these six errors were intended to be Gaze + Pinch but mistaken for a Lasso, and one was the inverse. We report on the mean proportion of selections for which participants chose each technique (Fig. 11c). When selecting groups of objects, participants overwhelmingly decided to use the Lasso technique (93% of selections), and also somewhat preferred Lasso to Gaze + Pinch for compound objects (66.67% of selections). This supports our third hypothesis, and suggests that users correctly understood the conditions under which the Lasso technique was most useful.

6.6.4 Qualitative Feedback. In the final survey, almost every participant rated the Hybrid technique to be their first choice (Fig. 12), and Lasso received two more second-choice votes than Gaze + Pinch. In free response feedback, many participants expressed an appreciation for the Hybrid technique’s flexibility across situations. For example, Participant 4 said “Lasso was more accurate than Gaze

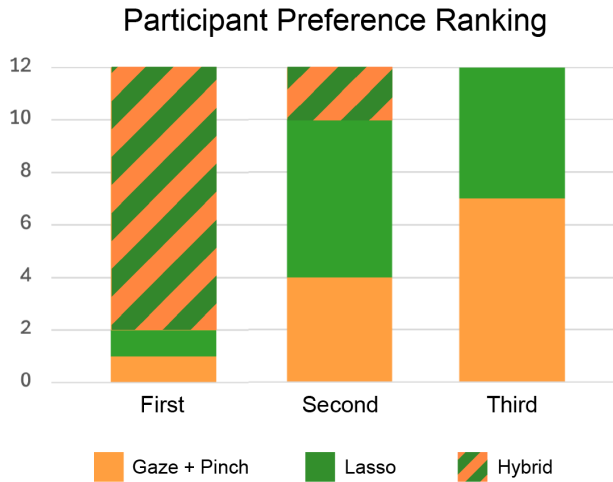


Figure 12: Results of our final questionnaire asking users to rate the three techniques by preference.

+ Pinch, but I appreciated being able to select between the two in the hybrid format because it was easier to gaze and pinch in unambiguous scenarios.” This supports our fourth hypothesis and the viability of this hybrid technique.

7 Discussion

The value of Squiggle’s lasso input is twofold: it provides drawing data that can help predict the intended target’s shape and size, and it often evokes expressive natural gaze behavior from the user. In our offline SAM2 experiments and our evaluation study, this rich input led to significantly improved accuracy when selecting compound objects and groups of objects. Moreover, our evaluation study results show that these benefits can be realized in a manner that is compatible with Gaze + Pinch selection, so users can conserve their effort with a quick pinch when the additional accuracy of a lasso is not required. Squiggle gives users increased agency over their in-world referencing, and our exploration of it demonstrates the power of multi-modal techniques where certain signals (in this case gaze movement) are evoked through specific gestures. This type of interaction is relatively unexplored and could be particularly useful for systems with many sensors but no visual display.

7.1 Benefits of Display-free Interaction

While continuous visual feedback is important to many interaction techniques, the present work suggests that not providing such feedback can also have advantages in some cases. We expect that the natural gaze behavior exhibited by users when performing an invisible lasso selection would have been less valuable if the cursor had been visible, as the cursor feedback itself may have drawn the user’s eye gaze. Further studies would be needed to confirm this, but similar effects were observed for Lystbæk et al.’s Gaze&Hand technique [35]. Mayer et al. [38] also found that the presence of a visible cursor slowed users’ target acquisition times compared to having no cursor. A further advantage of a display-free technique is that it may make a hybrid approach with Gaze + Pinch more

intuitive, as users can always gaze directly at their target without concern for where that will place their drawing cursor.

7.2 Stroke-Drawing Input

An area of research that has many similarities with Squiggle is that of location-independent stroke-drawing input [13, 33, 36, 47, 55, 59]. Many techniques have been developed for classifying a user’s drawing within a set of possible input shapes, which can then be associated with digital commands. Like Squiggle, this input is purely based on relative movement, and often it does not assume that the user can see their drawing. Given that stroke input could be supported by the same technology used for Squiggle, future work could explore how elements of these two techniques could be combined. For example, specific shapes could be drawn alongside a selection (e.g., reference something with Squiggle, then draw a checkmark to save it or a question mark to learn more) or as a special form of lasso (e.g., draw a heart around a target to indicate a preference for it). It would be interesting to see how other real-world invisible drawing tasks influence user’s gaze behavior. For example, the gaze evoked by performing a “highlighting” drawing gesture may be sufficient to select specific portions of text from the real world.

7.3 Future Improvements

We have identified a few potential ways to improve Squiggle from the version evaluated in this work. First, our prototype sizes all drawing input using the same generic linear function. Our data analysis indicated that a personalized scaling rule per-user would likely result in more accurate predicted drawing sizes, so adding a calibration process for this could be useful. Furthermore, there are additional drawing features we could potentially leverage in future work. For example, distinctive corners or protrusions of a drawing could be compared to edges in a scene, to help with difficult cases or as a mitigation for inaccurate gaze tracking. Finally, both our baseline and our drawing models were fine-tuned for all target hierarchy labels at once. However, since we’ve observed that users preferred to use Gaze + Pinch for simple objects and Lasso for compound objects and groups, this information could be incorporated into the training of those models in the future, so that they are each purpose-built to reference certain kinds of targets.

7.4 Limitations

Our goal with this work was to show the potential of multi-modal in-world lasso selection as an input technique, and we believe we have done this with our explorations using SAM2 and our evaluation study. However, the present work has several limitations. First, though our data collection included several virtual locations and a range of targets, it completely lacked any real-world data. It’s very likely that better models for target prediction could be developed for both of our input techniques with a larger and richer training set. We also did not explore the impact of environmental factors like lighting, occlusion, and background clutter, which could impact end-to-end performance because Squiggle relies on camera data.

Since we were working in a lab, our evaluation study featured a limited set of environments and targets. It would be useful to verify Squiggle’s performance in a wider range of scenarios. It would also

be interesting to compare Squiggle against other potential methods of expressive object referencing, like drawing in the image plane [42] or capturing a window of gaze data while the user examines their target [2]. Finally, our evaluation study was relatively small, with an n of 12, and short, lasting only two hours. Further research exploring Squiggle's performance "ceiling" with more practice, as well as its usefulness in specific applications, would be valuable.

8 Conclusion

In this work we introduced Squiggle, a real-world adaptation of the classic lasso selection technique. We found that even without visual feedback, participants can draw midair lassos consistently enough that this data can predict their target's dimensions, and that asking users to draw an object's outline can evoke useful natural gaze data. In an evaluation study we showed that Squiggle can help users reference targets (particularly compound objects and groups of objects) more effectively than they can with a single point of gaze input, and we provided support for a hybrid system that enables lasso selection alongside a Gaze + Pinch baseline. Squiggle expands user's capabilities for expressive in-world referencing, thereby expanding the possibilities for digital interactions with the physical world.

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